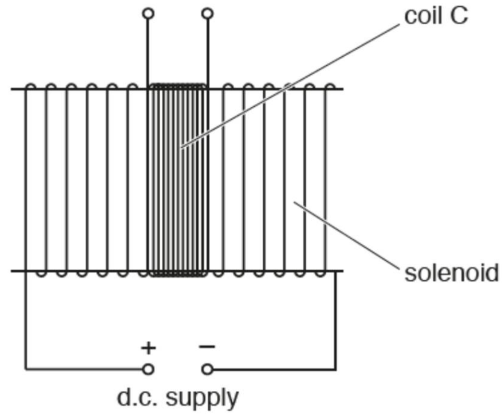
24 A solenoid has a coil C of wire wound tightly about its centre as shown.



Coil C has 96 turns.

The uniform magnetic flux ϕ (in weber) in the solenoid is given by the expression

$$\phi = 6.8 \times 10^{-6} \times I$$

where I is the current (in amperes) in the solenoid.

What is the average e.m.f. induced in coil C when a current of 3.3 A is reversed in the solenoid in a time of 2.4 ms?

A 0.0094 V

B 0.019 V

C 0.90 V

D 1.8 V

25 The diagram shows a thin semicircular metal wire MN of radius *r* falling through a region